

Name of the Experiment : Verification of half adder circuit using X-OR and AND gate.

Objective: The objective of this experiment is a half adder can be implemented by using X-OR and AND gate.

Theory:

Half adder: Half adder is a combinational logic circuit with two inputs and two outputs. Half adder is designed to add two single bit binary numbers. This single bit adder can be easily implemented with the help of X-OR gate for the output sum and an AND gate for the carry output.

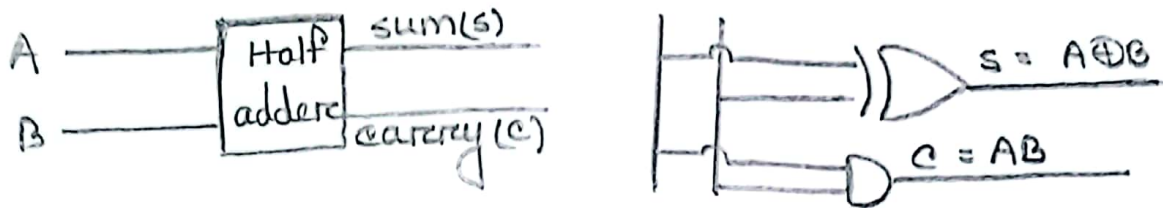


Fig: Half adder using X-OR and AND gate

Truth Table:

Input		Output	
A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$$\text{Sum} = \bar{A}B + A\bar{B} = A \oplus B$$

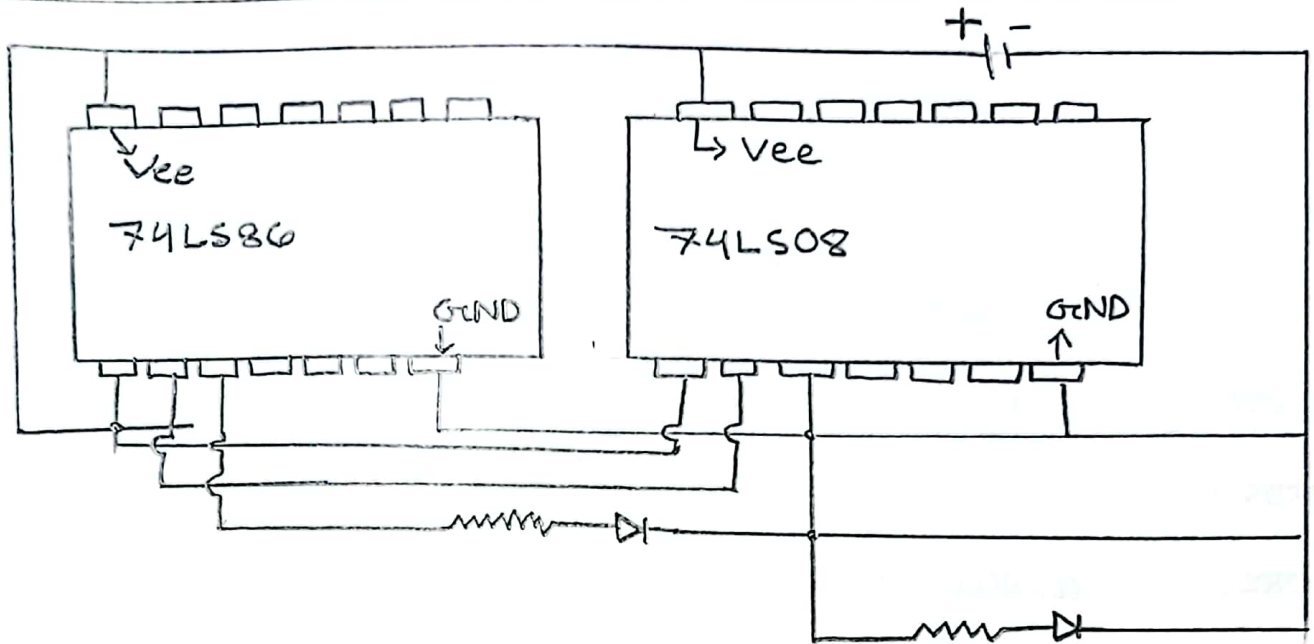
$$\text{Carry} = AB$$

Apparatus:

- i) AND-gate
- ii) X-OR gate
- iii) Bread Board
- iv) Resistor
- v) Power supply.
- vi) Wire
- vii) LED.

Procedure:

At first, we have to identify the pin numbers of the IC and then set it on the bread-board. Connect all the Vee with the positive of power supply and all the ground with negative and then connects the inputs and outputs wire with resistor and LED.



Result :

A	B	Sum	LED	Carry	LED
0	0	0	OFF	0	OFF
0	1	1	ON	0	OFF
1	0	1	ON	0	OFF
1	1	0	OFF	1	ON

Discussion :

In this experiment, we have checked all the gates of IC and found the active gates. We have found out the positions where the IC's get power supply. And we connected 7 no. in the GND and 14 No. pin in the Vcc. Then we connected all the wires in the exact point. When we worked in the trainer board. We keep the switch off state. Thus we successfully done the experiment.

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